Q06: Generalization, Memorization, and Emergence

Turn competing explanations into tests

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The question

Q06
What are models computing? Generalization vs. memorization? How do capabilities emerge with scale?

Generalization needs a target distribution

- Random held-out examples
- New wording / format
- New combinations of known skills
- New domain or causal regime

These are different claims, not interchangeable test sets.

Memorization and generalization can coexist

- Retaining examples is not automatically useless.
- A model can reuse fragments and still transfer.
- Benchmark shortcuts can look like competence.

The question is: what changes under a split designed to break the shortcut?

What is being computed?

- The direct output is a conditional distribution.
- Internal representations may encode reusable regularities.
- Behavior, intervention, and mechanistic evidence support different strengths of claim.

Few-shot behavior

- Prompt examples change outputs without changing weights.
- Success can reflect a useful context-sensitive rule—or a familiar template.
- Hold out compositions and paraphrases to distinguish them.

Brown et al. (2020)

Emergence: claim carefully

- Some benchmarks show threshold-like scale curves.
- A threshold can reflect a new computation, a metric, a prompt, or a mixture.
- Plot continuous metrics and disclose protocol/variance.

Wei et al. (2022); Schaeffer et al. (2023)

Toy held-out-combination test

- Train: most color–shape pairs.
- Test: withheld diagonal pairs.
- Lookup baseline fails unseen pairs.
- Rule solver succeeds if it captures the specified relation.

This isolates one notion of compositional generalization.

Takeaway

Evaluation principle
Name the target distribution and construct cases on which your intended explanation and its shortcut predict different outcomes.